EXP-1

EXPT NO: 1 VERIFICATION OF KIRCHHOFF'S LAWS

sample_EXP-1.pdf

AIM

a. To verify Kirchhoff's Voltage Law (KVL) for the given circuit. b. To verify Kirchhoff's Current Law (KCL) for the given circuits.

APPARATUS REQUIRED:

S.No. Components Range Quantity 1 Resistor 1kΩ 3 2 Voltmeter (DC) 0-30V 3 3 Ammeter (DC) (0-200)mA 3 4 Bread Board 1 5 Regulated Power Supply (0-30)V 1 6 Connecting wires As required

THEORY:

KVL: Kirchhoff's voltage law states that the sum of the voltage differences around any closed loop in a circuit must be zero. A loop in a circuit is any path that ends at the same point at which it starts. KCL: Kirchhoff's Current Law (KCL) Kirchhoff's Current Law states that the algebraic sum of the currents entering and leaving a node is equal to zero. By convention, currents entering the node are positive, and those leaving a node are negative

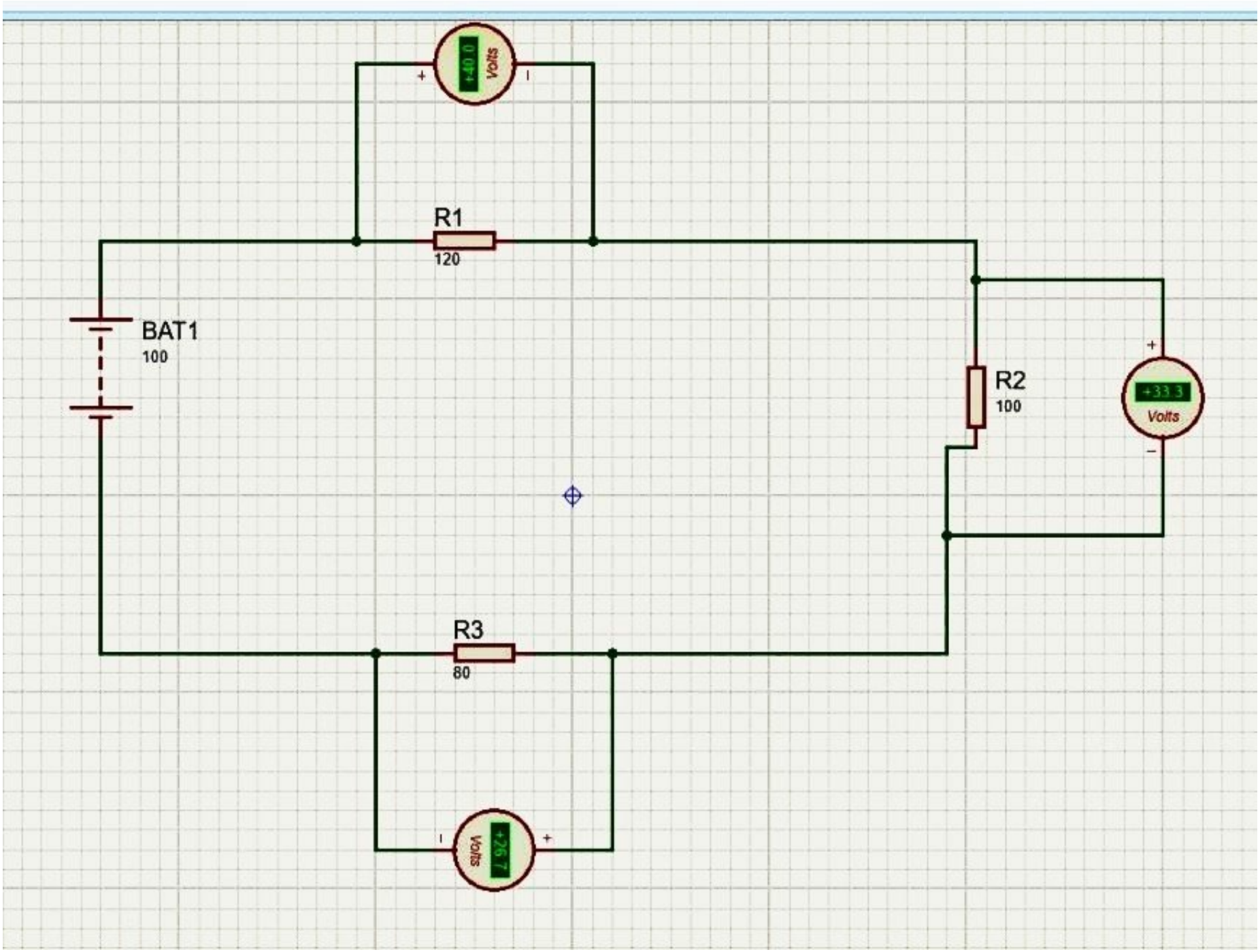
PROCEDURE:

a. KVL:

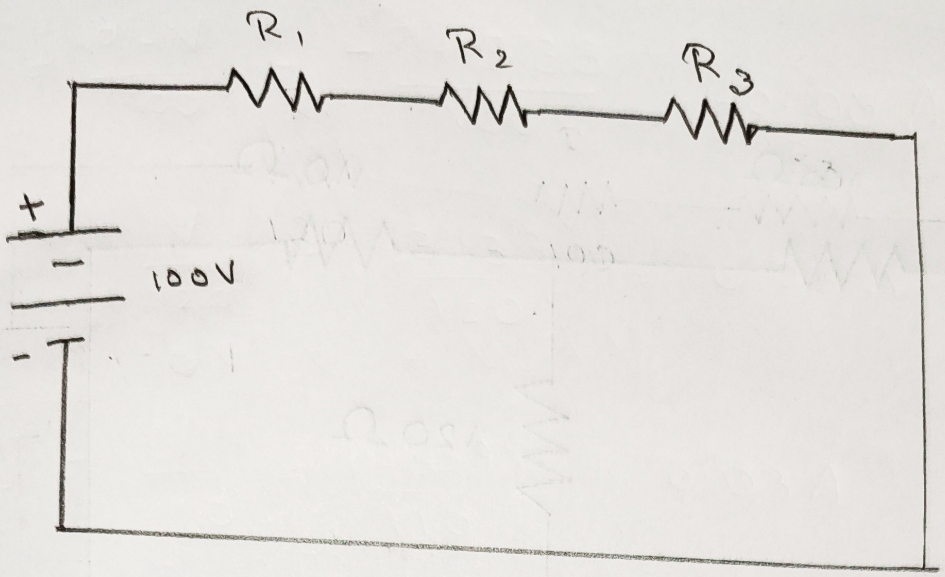
1. Connect as per the circuit diagram.
2. Check if the RPS voltage is set to zero voltage.
3. Check all the meters for null position.
4. Switch on the RPS.
5. Set the input voltage to a value between 0V to 30V.
6. Record the voltage values shown in the voltmeter connected across each resistor.
7. Take readings for different values of input voltage and tabulate them.

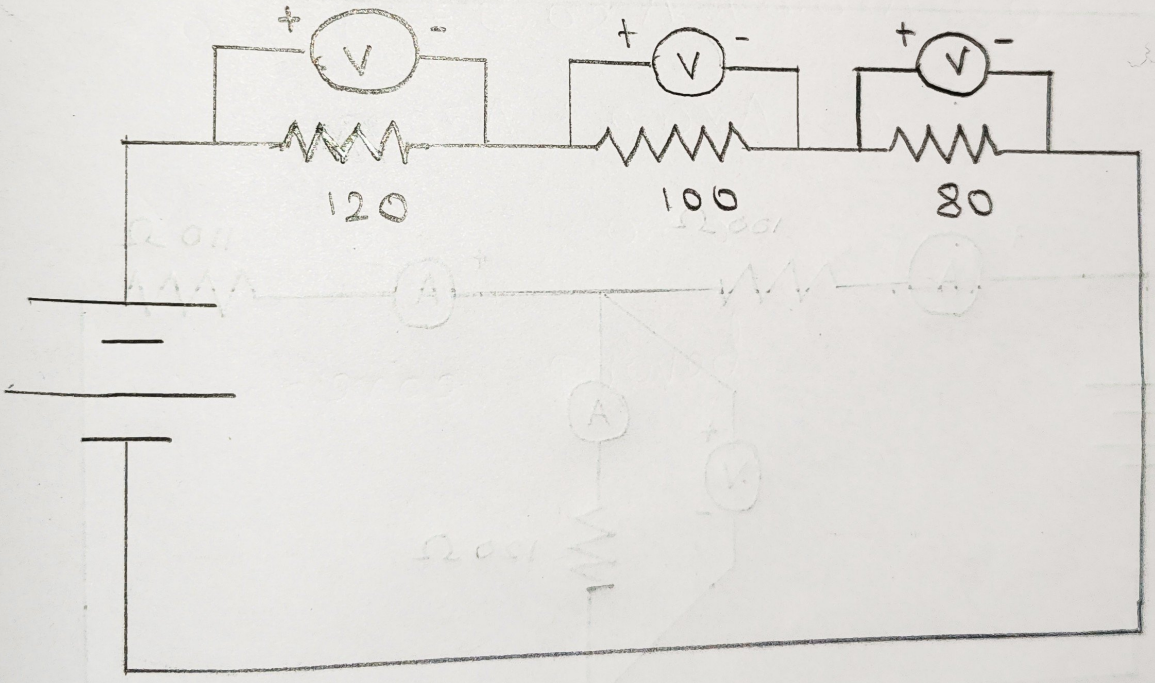
b. KCL:

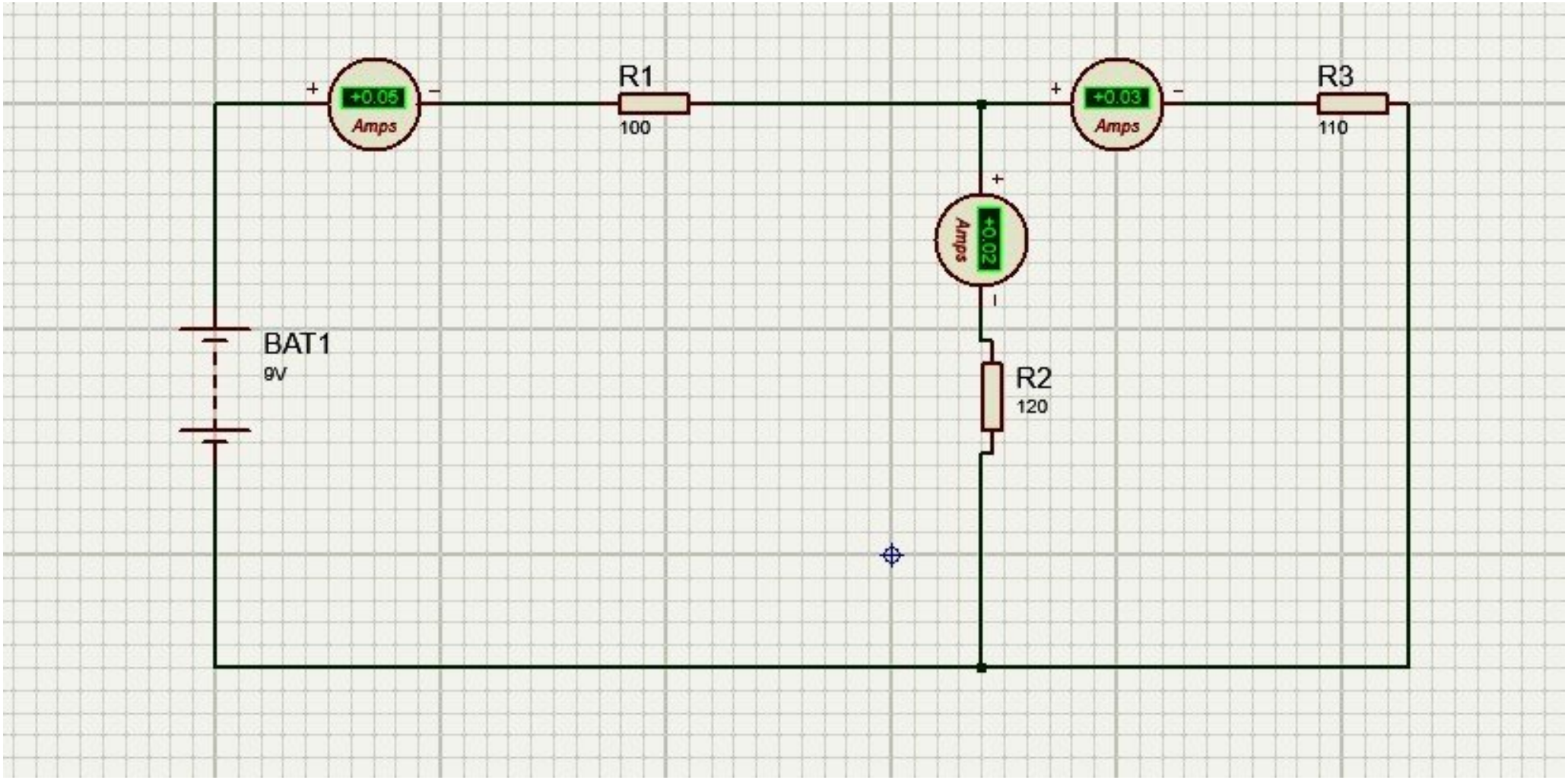
1. Connect as per the circuit diagram.
2. Check if the RPS voltage is set to zero voltage.
3. Check all the meters for null position.
4. Switch on the RPS.
5. Set the input voltage to a value between 0V to 30V.
6. Record the voltage values shown in the ammeter connected to each resistor.
7. Take readings for different values of input voltage and tabulate them.


+40.0
Volts
+
-
R1
120
BAT1
100
R2
100
+
+33.3
Volts
-
R3
80
-
+26.7
Volts
+

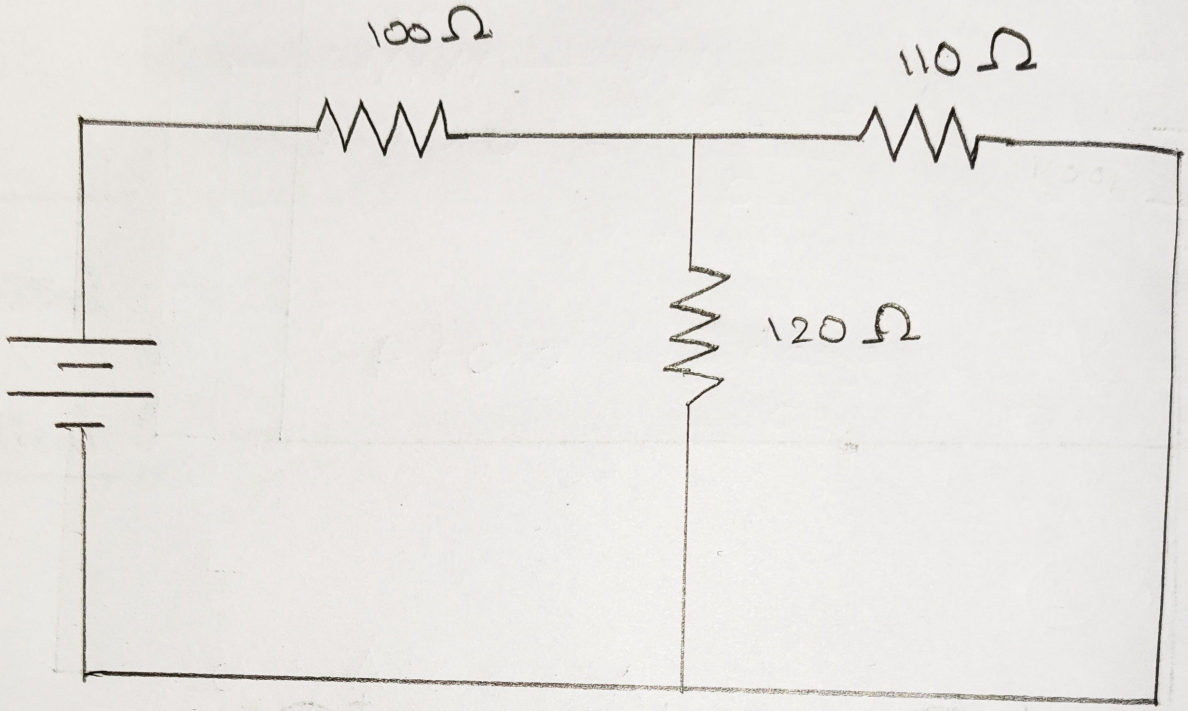
KVL

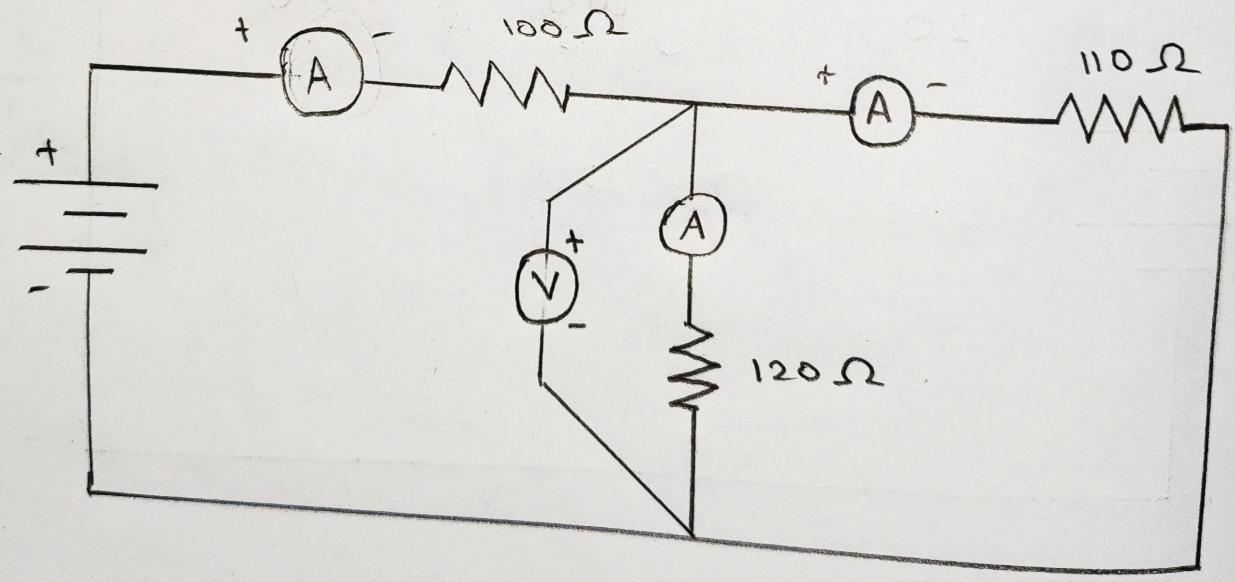





+0.05
Amps
R1
100
+0.03
Amps
R3
110
+0.02
Amps
BAT1
9V
R2
120

KCL





Kirchoff's Voltage law:

In loop 1 by KVL:

$$V = V_1 + V_2 + V_3$$

$$100 = IR_1 + IR_2 + IR_3 \rightarrow (1)$$

$$R_{eq} = R_1 + R_2 + R_3$$

$$R_{eq} = 120 + 100 + 80$$

$$R_{eq} = 300\Omega$$

$$I = \frac{V}{R_{eq}} = \frac{100}{300} = 0.333\ A$$

$$V_1 = IR_1 = 0.333 \times 120 = 39.9999\ V$$

$$V_2 = IR_2 = 0.333 \times 100 = 33.3\ V$$

$$V_3 = IR_3 = 0.333 \times 80 = 26.64\ V$$

Sub V_1, V_2 and V_3 values in (1)

$$100 = 39.99 + 33.4 + 26.64$$

$$100 = 100\ V$$

Hence Proved.

Kirchoff's Current Law:

In a junction, incoming current is equal to outgoing current.

At node A

$$I_1 + I_2 + I_3 = 0 \rightarrow (2)$$

$$\frac{V-9}{100} + \frac{V}{120} + \frac{V}{110} = 0$$

$$V\left[\frac{1}{100} + \frac{1}{120} + \frac{1}{110}\right] - \frac{9}{100} = 0$$

$$V\left[0.01 + 8.33 + 9.09\right] = 0.09$$

$$V\left[0.027\right] = 0.09$$

$$V = 3.333\,V$$

$$I_1 = \frac{V-9}{100} = \frac{3.333-9}{100} = -0.056\,A$$

$$I_2 = \frac{V}{120} = \frac{3.333}{120} = 0.0277\,A$$

$$I_3 = \frac{V}{110} = \frac{3.333}{110} = 0.03\,A$$

$$I_1 = \frac{9 - V}{100} = \frac{-9 + 3.33}{100} = -0.05\ A$$

$$I_2 = \frac{V}{120} = \frac{3.33}{120} = 0.02\ A$$

$$I_3 = \frac{V}{110} = \frac{3.33}{110} = 0.03\ A$$

At eqn (2)

$$-0.05A + 0.02A + 0.03A = 0$$

$$-0.05A + 0.05A = 0$$

$$0 = 0$$

∴ Hence proved.

KVL	Source (V)	V_1 (V)	V_2 (V)	V_3 (V)	$V_1 + V_2 + V_3$
Theoritical	100V	39.99V	33.4V	26.64V	100.03V
Practical	100V	39.99V	33.4V	26.64V	100.03V

KCL	I_1 (A)	I_2 (A)	I_3 (A)	V_A (V)
Theoritical	-0.05A	0.02A	0.03A	3.33V
Practical	-0.05A	0.02A	0.03A	3.33V

Result:

Thus, for the given circuit, Kirchhoff's laws a) KVL and b) KCL are proved.